

Jihwan Choi

📍 Seoul ✉ jhwan333@gmail.com ☎ 010-9692-6526 in jihwan-choi 🌐 jhwanchoi

Profile

A software engineer who designs and operates an MLOps inference & evaluation platform for autonomous-driving AI. I co-design the shared Kubernetes GPU cluster (leading scheduler & queue policy) and hold sole ownership of the inference/evaluation workflow (SVEvalFlow: PRISM, Inferit, MTBF Validation).

I cut a 20,000-hour-scale validation that previously took 6+ months on a single machine down to 1.5 months via distributed/parallel processing (173,101 clips, 99.97% success rate), and led a cost-driven On-Premise transition that saved ₩448M (55%) vs AWS. I built a log-analysis system processing 80M daily events that prevented ~₩100M/month in overbilling—consistently turning technical solutions into measurable business value. Most recently I build AI-native developer tools (LLM Wiki, Multi-Model Debate, Bedrock cost observability) and evangelize them across the organization.

Education

BS **Handong Global University**, Electronic Engineering, Computer Science Feb 2018 – Feb 2022

- GPA: 3.3/4.5
- **Coursework:** Data Structure, Algorithm, Computer Architecture, Deep Learning, Machine Learning

Experience

Stradvision, Software Engineer Seoul, South Korea
July 2022 – Present

MLOps Platform — Shared GPU Cluster Co-Design & SVEvalFlow (Inference/Eval) Owner (Oct 2025 – Present)

- Co-designed and built a shared GPU cluster on RKE2 Kubernetes (60+ heterogeneous GPUs: RTX 3090/4080/4090/5090, RTX 6000); **led scheduler & queue policy design**, integrated 15+ MLOps components (Ray/KubeRay, MLflow, Airflow, ArgoCD, KServe, Harbor)
- Collaborate on the training workflow (SVDeepFlow); hold **sole ownership of the inference/evaluation workflow (SVEvalFlow)** — Inferit, MTBF, and PRISM below
- Built 3 CI/CD pipelines (GitHub Actions + ArgoCD) and Scale-to-Zero serverless inference (KServe/Knative)
- **Tech:** RKE2, Ray, KubeRay, KServe, Knative, MLflow, Airflow, ArgoCD, Harbor, lakeFS, GitHub Actions, Helm

Inferit — svnet3 Build/Inference Acceleration & Artifact Management

- Built version/config/artifact management for building svnet3 (in-house autonomous-driving vision model) across many configs
- Accelerated single-machine inference into distributed/parallel execution across cluster GPUs — cut a 20,000-hour validation from an estimated 6+ months to 1.5 months
- Versioned customer option files (e.g., camera parameters), eliminating the 1–2 days of engineer-PM search/communication previously required
- 5+ major release versions supported; automated inference per customer issue ticket; integrated Evaluation Driven Development (eval automation)
- **Tech:** Ray, MLflow, lakeFS, Docker, Kubernetes

MTBF Validation — Large-Scale svnet3 Reliability Validation

- Runs svnet3 inference on customer-ODD driving data before/after release, detecting FP & AEB flags with automated evaluation; processed 173,101 clips at 99.97% success rate, with 205% higher throughput and 67.3% lower processing time vs a single machine
- Proposed and executed a cost-driven Cloud to On-Premise transition: AWS ₩820M to On-Premise ₩370M (55% saved); ran GPU hardware benchmarks & spec proposals, leveraged HCP Object Storage

- Automation enabled large-scale validation (previously feasible only at small scale) to be operated by a single person; Visual Inspection UI + Gemini-assisted review (human makes the final call)
- **Tech:** FastAPI, Airflow, Kubernetes, PostgreSQL, HCP Object Storage

PRISM — Customer Jira-Integrated ISP Conversion & Re-simulation Pipeline (Nov 2025 –)

- **Phase 1 (deployed/operating):** Event-driven ISP conversion via Jira Webhook to SQS to Lambda; in-house ISP conversion tool integration; ~150 requests/month, 80% faster than manual
- **Phase 2 (Resim, in dev):** Did the initial development, now technical lead supporting/managing a developer from another org; dual-version inference (issue + latest) via SWIT Docker Images, Airflow DAG trigger
- **Tech:** FastAPI, PostgreSQL, AWS (SQS, Lambda), Terraform, Airflow

Data & Labeling Automation (2022 – 2025)

- **WLS:** Analyzed 80M daily labeling-log events to prevent ~\$100M/month overbilling (settlement-audit confirmed); pinpointed bottlenecks feeding UI/UX and the ALAS model (+20% labeling efficiency) (Django, MongoDB, AWS SQS)
- **ALAS:** Auto-labeling microservice, reduced manual annotation workload by ~35% (FastAPI, AWS ECS/EKS, CDK)
- **IGS:** External-customer data ingestion service planning & architecture design (design-only) (Django, Airflow, Kubernetes)
- **SURF Compass:** Device monitoring service planning & design (design-only) (FastAPI, PostgreSQL)
- **RNS:** Automated release-note generation, 50% faster authoring (4h to 2h) (NestJS, Atlassian API)
- **SLT:** 2D LD labeling tool, improved productivity by 30% (C++/MFC)

Korea Institute of Robotics & Technology Convergence, Researcher

Pohang, South Korea

- Developed real-time gesture-based Human-Robot Interaction system
- Supported autism diagnosis assistance robot project

Jan 2022 – July 2022

Skills

Languages: Python, C++, TypeScript

Backend: Django, FastAPI, NestJS

Database: PostgreSQL, MongoDB, Redis

Cloud: AWS (EC2, EKS, ECS, SQS, SNS, Lambda, S3, Bedrock)

Infrastructure: Kubernetes (RKE2), Docker, Terraform, CDK, Rancher

ML/MLOps: Ray, KubeRay, MLflow, KServe, Knative, lakeFS

Workflow: Airflow, Celery

CI/CD: ArgoCD, GitHub Actions, Harbor, Helm

Monitoring: Grafana, Prometheus, Loki, DataDog

AI/LLM: Claude Code, Gemini CLI, Codex, MCP, AWS Bedrock, n8n

Tools: Git, Jira, Confluence, Atlassian API

AI-Native Development & Internal Evangelism

- **LLM Wiki:** Designed a Git + Markdown MLOps ops-knowledge system (natural-language authoring, search, Q&A, quality lint, sync); turned scattered individual ops knowledge into a team asset
- **Multi-Model Debate Hooks:** Cross-review workflow (Claude draft to GPT critique to Gemini review to Claude synthesis) raising reliability for design/document reviews (github.com/jhwanchoi/dotfiles)
- **Bedrock observability:** Monitoring infra (CloudWatch + Slack) for in-house LLM usage/cost visibility; cluster-ops Claude plugin (diagnostics, storage)
- **MTBF Visual Inspection AI assist:** Gemini-aided review (humans decide) reducing repetitive verification load; n8n automation prototypes (PR/news/cluster/recruiting assist)
- Hosted internal AI-driven development meetings; evangelized modern AI dev tools (Claude Code, Gemini CLI, Codex; Plugin/MCP/Skill)
- **Tech:** Claude Code, Gemini CLI, Codex, MCP, AWS Bedrock, CloudWatch, n8n